

ASSIGNMENT NO.6: Data Manipulation and Table Management

QUESTION 1

Q1) Create a MySQL database named CollegeDB and create two tables: Course and Student. Course Table Column Name Data Type Constraint CourseID INT PRIMARY KEY CourseName VARCHAR(50) NOT NULL Fees DECIMAL(10,2) NOT NULL Student Table Column Name Data Type Constraint StudentID INT PRIMARY KEY StudentName VARCHAR(50) NOT NULL Age INT City VARCHAR(30) CourseID INT FOREIGN KEY

Query

```
CREATE TABLE Course (  
    CourseID INT PRIMARY KEY,  
    CourseName VARCHAR(50) NOT NULL,  
    fees Decimal (10,2) NOT NULL  
);
```

```
CREATE Table Student (  
    StudentID INT PRIMARY KEY,  
    studentName VARCHAR(50) NOT null,  
    Age INT,  
    city VARCHAR(30),  
    CourseID INT,  
    FOREIGN KEY (CourseID) REFERENCES course(CourseID);
```

QUESTION2

Q2) Insert at least 4 records into the Course table and 6 records into the Student table. B.SC. DATA SCIENCE Make sure every student has a valid CourseID from the Course table.

QUERY

INSERT INTO Course (CourseID, CourseName, Fees)

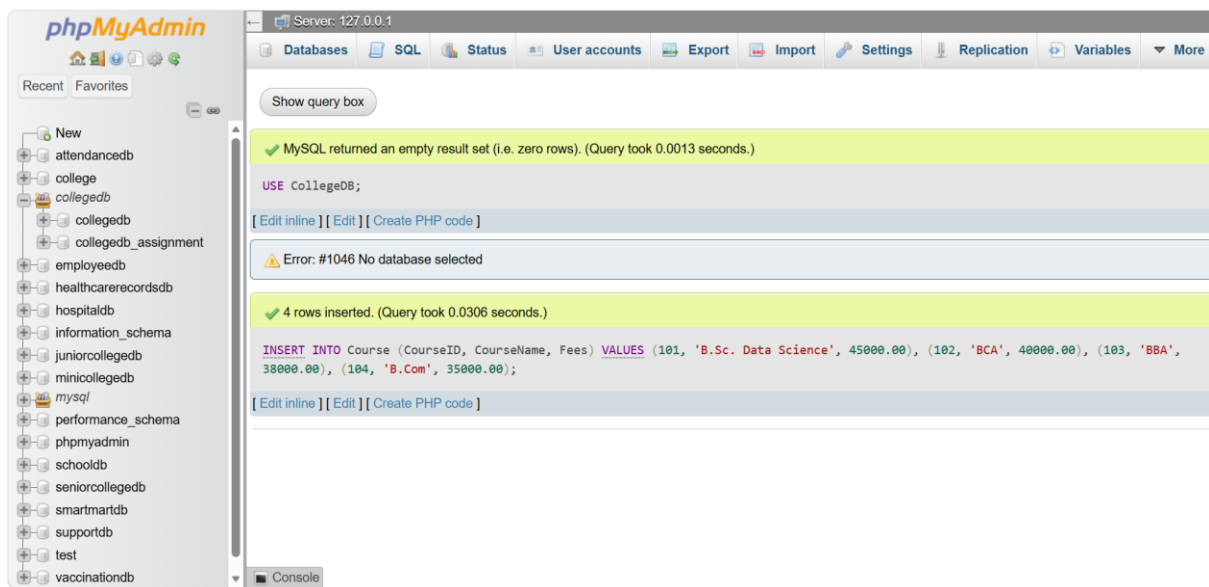
VALUES

(101, 'B.Sc. Data Science', 45000.00),

(102, 'BCA', 40000.00),

(103, 'BBA', 38000.00),

(104, 'B.Com', 35000.00);



INSERT INTO Student (StudentID, StudentName, Age, City, CourseID)

VALUES

(1, 'Aarav Sharma', 19, 'Pune', 101),

(2, 'Sneha Patil', 20, 'Nashik', 101),

(3, 'Rahul Joshi', 19, 'Mumbai', 102),

(4, 'Priya Shah', 21, 'Pune', 103),

(5, 'Rohan Deshmukh', 20, 'Nagpur', 104),

(6, 'Ananya Kulkarni', 19, 'Pune', 102);

The screenshot shows the phpMyAdmin interface with the 'collegedb' database selected. The 'Course' table is open, displaying the results of the query `SELECT * FROM Course;`. The table has three columns: **courseID**, **CourseName**, and **Fees**. There are four rows of data.

	courseID	CourseName	Fees
☐	101	B.Sc. Data Science	45000.00
☐	102	BCA	40000.00
☐	103	BBA	38000.00
☐	104	B.Com	35000.00

The screenshot shows the phpMyAdmin interface with the 'collegedb' database selected. The 'Student' table is open, displaying the results of the query `SELECT * FROM Student;`. The table has five columns: **StudentID**, **StudentName**, **Age**, **City**, and **CourseID**. There are six rows of data.

	StudentID	StudentName	Age	City	CourseID
☐	1	Aarav Sharma	19	Pune	101
☐	2	Sneha Patil	20	Nashik	101
☐	3	Rahul Joshi	19	Mumbai	102
☐	4	Priya Shah	21	Pune	103
☐	5	Rohan Deshmukh	20	Nagpur	104
☐	6	Ananya Kulkarni	19	Pune	102

QUESTION 3

Q3) Write a MySQL query to insert one new student record into the Student table. The record should include: Field StudentID StudentName Age City CourseID

QUERY

INSERT INTO Student

(StudentID, StudentName, Age, City, CourseID)

VALUES

(7, 'Vikram Singh', 20, 'Delhi', 101);

VERIFY:

Select * from student

The screenshot shows the phpMyAdmin interface for a MySQL server (127.0.0.1). The left sidebar displays a list of databases, including 'collegedb' and 'collegedb_assignment'. The main panel shows the 'SQL' tab with the query 'SELECT * FROM Student;' executed successfully, displaying 7 rows. The table structure is as follows:

StudentID	StudentName	Age	City	CourseID
1	Aarav Sharma	19	Pune	101
2	Sneha Patil	20	Nashik	101
3	Rahul Joshi	19	Mumbai	102
4	Priya Shah	21	Pune	103
5	Rohan Deshmukh	20	Nagpur	104
6	Ananya Kulkarni	19	Pune	102
7	Vikram Singh	20	Delhi	101

QUESTION 4

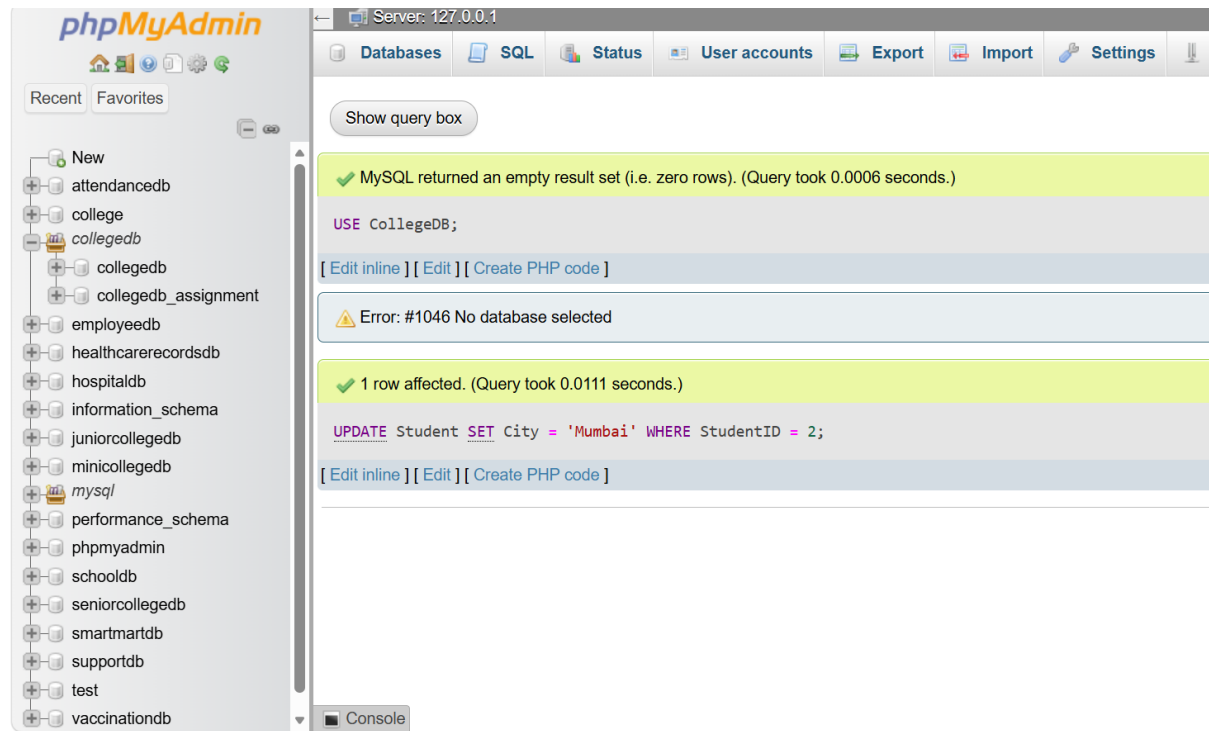
Q4) Write a MySQL query to update the city of a student using StudentID.
Example requirement: Change Required Update the city of StudentID 2 to Mumbai

QUERY

UPDATE Student

SET City = 'Mumbai'

WHERE StudentID = 2;



VERIFY:

Select * from student

Where studnetID =2

The screenshot shows the phpMyAdmin interface. On the left is a sidebar with a database list including 'collegedb'. The main panel displays an error message: 'Error: #1046 No database selected'. Below this, a green bar indicates 'Showing rows 0 - 0 (1 total, Query took 0.0009 seconds.)'. The SQL query entered is 'SELECT * FROM Student WHERE StudentID = 2;'. Below the query, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A table with 5 columns is shown: 'StudentID', 'StudentName', 'Age', 'City', and 'CourseID'. The first row contains the values: 2, Sneha Patil, 20, Mumbai, 101. Below the table are controls for 'Show all', 'Number of rows' (set to 25), and 'Filter rows'. At the bottom, there are buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

QUESTION 5

Q5) Write a MySQL query to update the fees of a course using CourseID.
Example requirement: Change Required Update the fees of CourseID 101 to 47000

QUERY

UPDATE Course

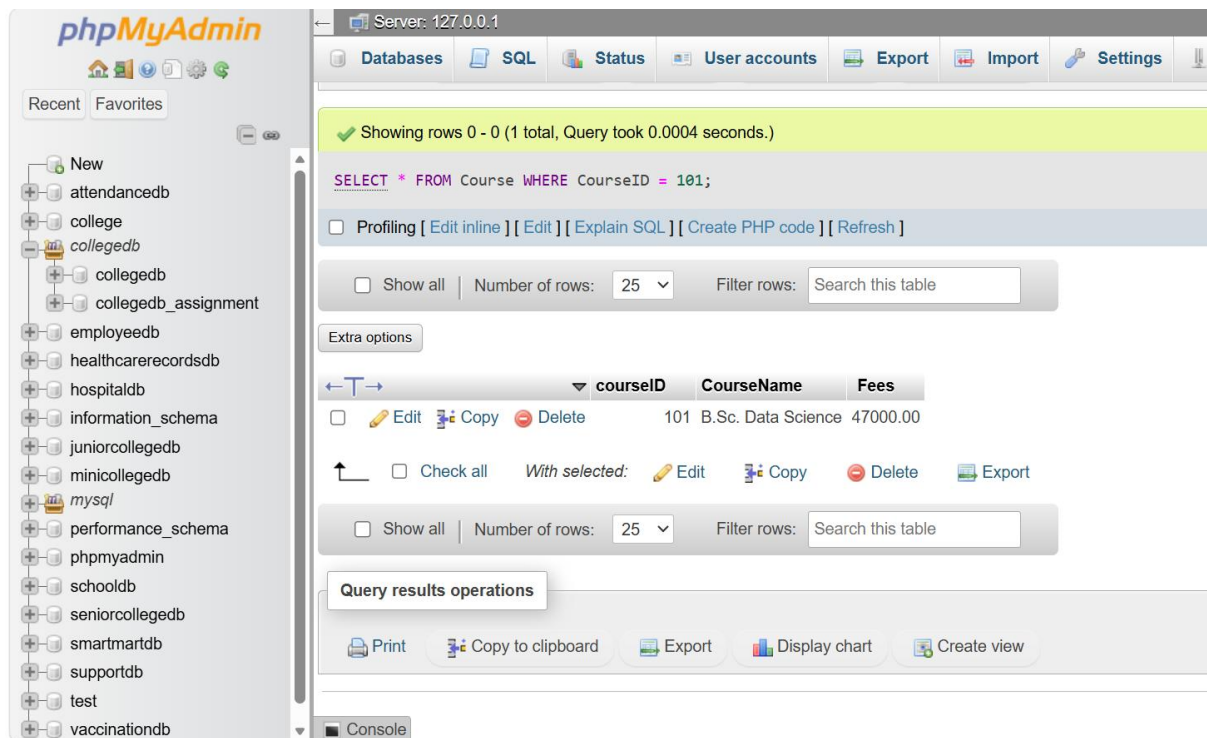
SET Fees = 47000.00

WHERE CourseID = 101;

VERIFY:

Select * from COURSE

Where studnetID =101;



QUESTION 6

Q6) Write a MySQL query to delete one student record using StudentID. Also write one line explaining why the WHERE clause is important while using DELETE.

QUERY

```
DELETE FROM Student
WHERE StudentID = 6;
```

VERIFY:

```
Select * from STUDENT
```

phpMyAdmin

Server: 127.0.0.1

Databases SQL Status User accounts Export Import Settings Replication

Select * from STUDENT;

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	StudentID	StudentName	Age	City	CourseID
☐ Edit Copy Delete	1	Aarav Sharma	19	Pune	101
☐ Edit Copy Delete	2	Sneha Patil	20	Mumbai	101
☐ Edit Copy Delete	3	Rahul Joshi	19	Mumbai	102
☐ Edit Copy Delete	4	Priya Shah	21	Pune	103
☐ Edit Copy Delete	5	Rohan Deshmukh	20	Nagpur	104
☐ Edit Copy Delete	7	Vikram Singh	20	Delhi	101

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Console

THE WHERE CLAUSE ENSURES THAT ONLY THE SPECIFIED STUDENT RECORD IS DELETED.

WITHOUT WHERE, ALL RECORDS FROM THE STUDENT TABLE WOULD BE DELETED.